

Shaswat Singh

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EXPERIENCE

AI Engineering and Integration

Sep. 2025 – Mar. 2026

AI Engineering Intern

- Responsible for designing, developing, and integrating AI solutions into existing systems and products.
- Built AI agents, connected APIs, deployed scalable AI pipelines/agents, and ensured seamless integration.
- Key project developed during the internship: **PrepAI** – an intelligent interview & notes assistant platform. [Certificate](#)

RESEARCH

AccidentGPT: Enhancing Road Safety Through Explainable AI and Real-Time Predictive Alerting | *Under Review*

- Proposed a proactive XAI framework for predicting road accident severity before occurrence, trained on 12,316 Indian police records (2017–2022) spanning 32 features across three severity classes.
- Benchmarked 11 classifiers; **Random Forest achieved 93.36% accuracy (macro F1 = 0.93)**; CNN baseline reached 94% but was 8× slower, making RFC the production choice for real-time edge deployment.
- Integrated **TreeSHAP** for per-instance explainability — top risk factors: vehicle count, time of day, junction type, weather, and casualty count — and built a **GPT-to-TTS alert pipeline** converting SHAP outputs into plain spoken warnings under 500 ms end-to-end latency.
- Keywords:** Explainable AI, SHAP, Random Forest, CNN, NLP, Road Safety, Real-Time Alerting, Indian Traffic

PROJECTS

PrepAI | *Intelligent Interview & Notes Assistant*

- Comprehensive, all-in-one AI platform designed to help job seekers and professionals master interview skills and deepen subject knowledge using conversational and generative AI technologies.
- Tech Stack:** FastAPI, SQLAlchemy, Pydantic, LlamaIndex, Vapi, Groq, Argon2, ChromaDB, React, TailwindCSS, TypeScript, Prompt Engineering

GUARDIAN-EYE.AI | *Real-Time Violence Detection System*

- Designed an AI and computer vision-based system for public safety in urban India; detects violent activities in real-time and alerts authorities via Telegram, enhancing traditional surveillance systems.
- Tech Stack:** Python, YOLOv8, Telegram API, OpenCV, Custom LLMs

DevEcho | *AI-Powered Social Media Assistant*

- Intelligent assistant that automates LinkedIn post generation from user-provided content using LLMs and Retrieval-Augmented Generation (RAG); integrated data collection pipelines for topical research.
- Tech Stack:** Python, LangChain, Groq, ChromaDB, Telegram API

EDUCATION

University of Lucknow

2022 – 2025

Bachelor of Computer Application | CGPA: 6.9

- Coursework:** Data Structures, Software Engineering, Algorithms, Database Management, Computer Networks, Operating System, Artificial Intelligence, Machine Learning, Computer Architecture

CERTIFICATES

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate

ACHIEVEMENTS

Hackofiesta 6.0 – AISpire UP Hackathon

IIIT Lucknow

Finalist

- Selected as a finalist at one of UP's premier AI-focused hackathons, hosted at IIIT Lucknow.

SKILLS

Programming Languages : Python

Frameworks : Hugging Face, LangChain, Llama Index, Unsloth, Transformers, FastAPI, NumPy, Pandas, Seaborn, PyTorch, vllm, LM-Deploy, SQLAlchemy, Pydantic, Argon2, Vapi, llama-cpp, TRL, OpenAI, GRPO, SFT, imblearn, sklearn

Development Tools : Git, Ollama, Ngrok, Jupyter Notebook, Vercel, Docker

Databases : ChromaDB, PostgreSQL, Qdrant

Fullstack : FastAPI, Argon2, SQLAlchemy, Pydantic, React, TypeScript, Tailwind CSS, Express